BibleGoAI: AI-Driven Synthesis of Biblical Commentary Using Multi-Stage LLM Pipelines and Retrieval-Augmented Generation

Eric Hao Brandon Xia Esther Faulk Sam Faulk Mark Miller
James Reinarz David Kuo Sarah Kuo Sam Kuo Philip Tsao
Samuel Huoh Regina Graphenreed Keith Dolcy Rex Beck
https://www.biblego.org/

Abstract

BibleGoAI is a commentary-synthesis pipeline that automatically distills multi-author biblical commentary into concise, source-adherent summary paragraphs—called *Brief Helpful Texts* (BHTs)—from a curated corpus of fourteen commentators spanning four centuries. The pipeline was run once over the full Bible to produce a fixed BHT corpus; each BHT targets a 30–80 word paragraph (observed means 63–84 words across the three generation variants reported in this paper), with automated quality scoring and provenance tracking via footnotes linking every claim back to its source commentary. The pipeline combines large language models (LLMs), retrieval-augmented generation (RAG), and semantic similarity scoring across a multi-stage flow: quote extraction, tiered synthesis, constraint-enforced generation with best-of-five selection, and cross-encoder footnote linking. We evaluate the generated corpus across 7,957 New Testament verses and the full 39-book Old Testament, comparing a single-pass regular pipeline against a two-stage split pipeline for OT books. The resulting BHTs are served to end users through BibleGo.org—a web application providing interactive commentary exploration, original-language word study, and full commentator browsing—reaching approximately 100–300 unique users per month. Beyond the engineering contribution, BibleGoAI is also a record of a volunteer-driven faith-community collaboration: commentator curation by ministry leaders, a 28-volunteer digitization effort of Henry Alford's 2,544-page New Testament commentary, and iterative usability testing with the project's Bible study community.

1 Introduction

The study of the Bible has generated one of the largest and most enduring bodies of scholarly commentary in human history. Prominent theologians—from John Calvin in the sixteenth century to Archibald T. Robertson in the twentieth—have produced detailed, verse-by-verse analyses that illuminate the linguistic, historical, and theological dimensions of the biblical text. For many Bible studiers, these commentaries are indispensable. Yet the sheer volume of material presents a practical challenge: consulting even a handful of commentators for a single verse requires navigating multiple lengthy texts, synthesizing sometimes divergent perspectives, and extracting the most relevant insights.

BibleGoAI grew out of a simple goal: *help people study the Bible using time-tested commentaries presented in a simple and accessible way*. The project was initiated and guided by a group of Christians—mostly living in the Cleveland, Ohio area—who have been ministering to churches for years. Through weekly round table meetings, this group of volunteers and full-time ministry leaders brainstormed how artificial intelligence could make quality biblical commentary more accessible, studied the needs of modern Bible readers, and analyzed existing platforms such as BibleHub, BlueLetterBible, and StudyLight to understand how commentary is currently delivered online. The original commentary texts were sourced from StudyLight.org.

Modern advances in natural language processing (NLP) and large language models (LLMs) offer new possibilities for addressing this challenge. Systems based on retrieval-augmented generation (RAG) have demon-

strated the ability to ground LLM outputs in authoritative source documents [Lewis et al., 2020], while sentence embeddings [Cer et al., 2018, Reimers and Gurevych, 2019] enable fine-grained semantic comparison between generated and source texts. These capabilities make it feasible to build systems that not only summarize but also *validate* the fidelity of their outputs against the original material.

In this paper, we present **BibleGoAI**, a multi-stage pipeline that transforms curated biblical commentary into concise, source-adherent summary paragraphs called *Brief Helpful Texts* (BHTs). The name reflects the project's core value: bringing the richness of difficult-to-read commentary to the fingertips of Bible readers in an accessible way. The system is designed around three principles:

1. **Fidelity**: BHTs must be grounded in the actual words of trusted commentators, not in the LLM's own knowledge. At least 30% of each BHT's words must originate from source quotes (with a target of 70% and an upper bound of 90%).
2. **Conciseness**: Each BHT targets 30–80 words in 1–3 sentences (observed means 63–84 words across the three generation variants), making it practical for quick reference during Bible study.
3. **Traceability**: Every claim in a BHT is linked back to its source commentary via footnotes with similarity scores, enabling readers to verify and explore further.

The pipeline was run once over the entire New Testament (7,957 verses across 27 books) and the full Old Testament (all 39 books), using two pipeline variants: a single-pass regular pipeline for 24 books and a two-stage split pipeline for 20 books (see Section 6 for details). It draws from a curated corpus of 14 commentators spanning four centuries and organized into three tiers of scholarly importance. The NT was publicly launched at a New Year's church conference in December 2023, with the OT following at a Memorial Day church conference in 2024. The resulting BHTs are served to end users through **BibleGo.org**, a web application that provides interactive commentary exploration, original-language word study via Strong's concordance, and full commentator browsing, serving approximately 100–300 unique users per month with monthly page-view volumes ranging from $\sim$200 to $\sim$2,500 clicks depending on seasonal and community activity.

Our contributions are as follows:

- A multi-stage pipeline for commentary synthesis combining LLM-based quote extraction, tiered prompt engineering, and constraint-enforced generation.
- A composite quality scoring framework that integrates semantic similarity and source fidelity checks.
- An OT-specific two-pipeline design—a single-pass regular pipeline and a two-stage split pipeline—that together covers all 39 Old Testament books, with per-pipeline metrics reported across the generated corpus.
- A user-facing application (BibleGo.org) serving the generated BHTs to a live Bible-study community, demonstrating the practical impact of the system.

Project context. BibleGoAI is a volunteer-driven collaboration among Christians—mostly based in the Cleveland, Ohio area—with backgrounds in software engineering, machine learning, and pastoral ministry. The project proceeded through weekly round-table meetings in 2023, a volunteer commentary digitization effort (Henry Alford's 2,544-page New Testament commentary, digitized by 28 volunteers), and an intensive April–May 2024 development sprint. The NT was publicly launched at a New Year's church conference in December 2023 and the OT at a Memorial Day church conference in 2024. Several technical decisions were shaped by this community context: the commentator corpus was curated by ministry leaders for theological reliability; usability testing was conducted with members of the project's Bible study community; the NASB 1995 translation was adopted after the originally requested ESV was denied by its publisher; and a sustained cadence of internal demos served as the primary feedback mechanism. These community and process constraints are a meaningful part of what this paper documents.

2 Related Work

Our work draws on several intersecting areas of research.

Abstractive and extractive summarization. Text summarization has been a long-standing goal of NLP, with approaches ranging from extractive methods that select key sentences [Devlin et al., 2019] to abstractive methods that generate novel text conditioned on source documents [Vaswani et al., 2017]. BibleGoAI employs a hybrid approach: the quote extraction phase is extractive (selecting verbatim passages), while the BHT generation phase is abstractive (synthesizing a coherent paragraph from selected quotes). This hybrid design allows us to

leverage the fluency of LLMs while maintaining explicit grounding in source material.

Retrieval-augmented generation. RAG systems [Lewis et al., 2020] address the hallucination problem inherent in LLMs by conditioning generation on retrieved documents. Our system applies RAG specifically to multi-verse biblical commentary where per-verse segmentation is impractical, and organizes the retrieved context into a tiered structure across multiple independent commentaries before generation—a design choice that differentiates it from typical RAG systems that retrieve from a single corpus.

Semantic similarity and sentence embeddings. The Universal Sentence Encoder [Cer et al., 2018] and Sentence-BERT [Reimers and Gurevych, 2019] provide dense vector representations that enable fine-grained semantic comparison. We use these embeddings for three distinct purposes: retrieval indexing (via FAISS [Johnson et al., 2019]), quality scoring (measuring BHT-to-commentary and BHT-to-verse similarity), and footnote generation (linking BHT sentences to source quotes via cross-encoder scoring).

AI in digital humanities and biblical studies. The application of NLP to religious and historical texts is a growing field. Prior work has explored machine translation of biblical texts [NLLB Team, 2022], named entity recognition in historical documents, and topic modeling across theological corpora. BibleGoAI contributes to this space by focusing specifically on *commentary synthesis*—a task that requires not only understanding the source texts but also respecting the scholarly authority and interpretive nuances of individual commentators.

Existing digital commentary platforms. Several widely used platforms make public-domain biblical commentary available online, including BibleHub, BlueLetterBible, StudyLight, and Logos Bible Software (commercial). These systems primarily serve as *browsers* over digitized commentary: a reader selects a verse and sees the corresponding excerpts from each commentator, one commentator at a time. They do not attempt to synthesize across commentators, enforce length budgets, or link generated text back to source quotes. BibleGoAI complements rather than replaces these tools: the source corpus itself is drawn from StudyLight, and BibleGo.org links out to full commentator text for readers who want to dive deeper. The contribution here is the *synthesis layer*—producing a short, source-grounded summary that a reader can consume in seconds rather than a multi-commentator reading session—together with the provenance mechanism that keeps each synthesized claim traceable to the underlying commentary.

3 System Architecture

BibleGoAI is composed of three major subsystems: (1) a data layer housing the biblical text and commentary corpus, (2) an AI pipeline that generates, scores, and validates BHTs, and (3) a web application that serves the results to end users. Figure 1 provides an overview of the end-to-end architecture.

3.1 Data Sources

The system operates on two primary data sources:

Biblical text. The New American Standard Bible (NASB, 1995 edition) provides the target verse text for all 66 books of the Bible. Each verse is stored as a JSON record with chapter, verse number, and text content. The NASB was selected for its reputation as a highly literal translation, which aligns well with the commentary corpus.

Commentary corpus. The commentary collection comprises 14 curated commentators spanning the 16th through 20th centuries (four centuries of biblical scholarship). For the New Testament, nine commentators form the active corpus: Henry Alford, Jamieson-Fausset-Brown, Albert Barnes, Marvin Vincent, John Calvin, Philip Schaff, Archibald T. Robertson, John Gill, and John Wesley. They are organized into three tiers (Table 1). For the Old Testament, twelve commentators are used: Jamieson-Fausset-Brown, Albert Barnes, John Gill, Keil & Delitzsch, Cyrus I. Scofield, Johann P. Lange, John Calvin, William Kelly, Charles H. Mackintosh, Charles A. Coates, Arthur W. Pink, and Henri L. Rossier, similarly tiered (Table 2). Four commentators (Jamieson-Fausset-Brown, Albert Barnes, John Gill, John Calvin) appear in both the NT and OT corpora, sometimes at different tiers.

All commentaries were curated and vetted by a group of Christians from the Cleveland, Ohio area, ensuring theological reliability and scholarly quality.

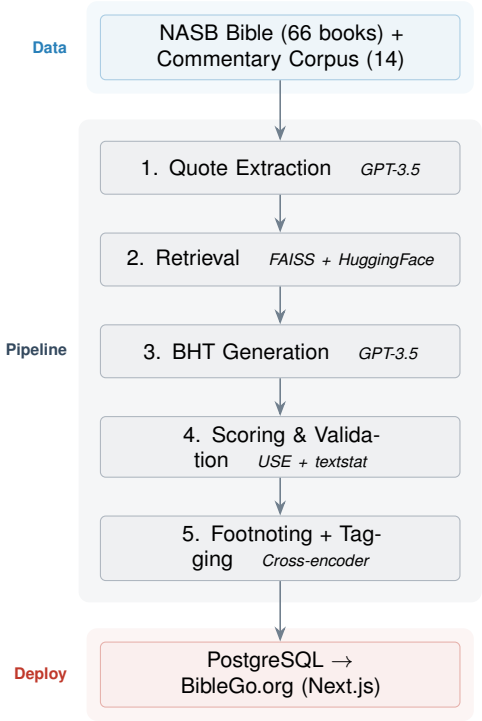

Figure 1: End-to-end system architecture. Commentary and Bible text flow through the five-phase AI pipeline, producing scored BHTs with footnotes that are stored in PostgreSQL and served via BibleGo.org.

3.2 Commentator Tiering

The tiering system serves a critical role in the generation process. When synthesizing a BHT, the LLM is instructed to draw *primarily* from Tier 1 commentators, *sparingly* from Tier 2, and *minimally* from Tier 3. This weighted approach ensures that the most trusted and comprehensive commentators dominate the output while allowing supplementary perspectives to enrich the synthesis. Tables 1 and 2 list the active NT and OT commentators with their tier assignments and approximate date ranges.

3.3 Prompt Engineering

The pipeline relies on three families of prompts, each iteratively refined through multiple versions and recorded in the project repository: **choicest** (per-commentator quote extraction), **BHT** (single-paragraph synthesis, with both a single-pass *regular* variant and a two-stage *split* variant), and **tag commentary** (verse-level indexing of raw commentary text used during preprocessing). A fourth family, *choicest range*, was developed to extract single-verse quotes from multi-verse commentary but did not reach the production pipeline; verse-range handling was instead addressed through retrieval-augmented generation (Section 4, Phase 2) combined with the split pipeline for narrative OT books. The full prompt specifications, including the word-count and quote-proportion constraints used in production, appear in Section 4, Phases 1 and 3.

Table 1: Active New Testament commentators and tier assignments. Tier assignments are defined per testament; several commentators (Jamieson-Fausset-Brown, Albert Barnes, John Gill, John Calvin) appear in both the NT and OT corpora and may receive different tiers.

Commentator	Era	Tier
Henry Alford	1810–1871	1
Jamieson, Fausset & Brown	1871	1
Marvin R. Vincent	1834–1922	1
Archibald T. Robertson	1863–1934	1
Albert Barnes	1798–1870	2
Philip Schaff	1819–1893	2
John Wesley	1703–1791	3
John Gill	1697–1771	3
John Calvin	1509–1564	3

Table 2: Active Old Testament commentators and tier assignments.

Commentator	Era	Tier
Keil & Delitzsch	1807–1890	1
Arthur W. Pink	1886–1952	1
Henri L. Rossier	1835–1928	1
Charles A. Coates	1862–1945	1
Charles H. Mackintosh	1820–1896	1
Jamieson, Fausset & Brown	1871	2
Johann P. Lange	1802–1884	2
William Kelly	1821–1906	2
Cyrus I. Scofield	1843–1921	2
Albert Barnes	1798–1870	3
John Gill	1697–1771	3
John Calvin	1509–1564	3

3.4 Commentary Digitization

Not all commentaries in the corpus were available in digital form. Two major digitization efforts were undertaken by volunteers to expand the commentary corpus:

Alford's New Testament for English Readers. Henry Alford's four-volume commentary (1868–1872)

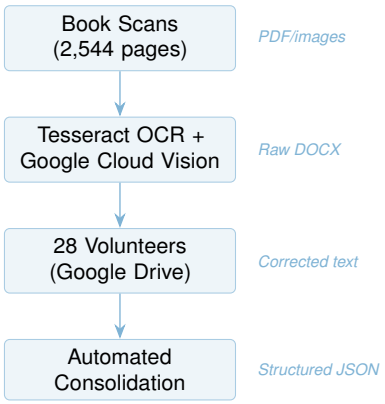

Figure 2: Commentary digitization pipeline used for Alford's and Govett's commentaries.

was digitized from physical book scans totaling 2,544 pages across all 27 New Testament books. The pipeline consisted of three stages: (1) *extraction* via Tesseract OCR and Google Cloud Vision, converting scanned pages to text; (2) *human correction*, in which 28 volunteers spent hundreds of hours reviewing and correcting OCR errors through a coordinated Google Drive spreadsheet; and (3) *automated consolidation*, which detected chapter boundaries, parsed verse-level commentary, performed spell-checking, and produced structured output files. This effort represents a significant scholarly contribution independent of the AI pipeline.

Govett's Commentary on Revelation. Robert Govett's commentary on Revelation was digitized through the same volunteer-driven process, spanning 627 pages and producing 274 individual verse commentary files covering all 22 chapters. Both digitization projects followed the same three-stage pipeline, demonstrating its replicability for future commentary digitization.

3.5 Original Language Tagging

A separate pipeline was developed to tag every word in the Bible with its corresponding Hebrew or Greek Strong's concordance number. The raw NASB text, with original-language tags provided courtesy of the NASB, was processed through a dedicated parser that produced two outputs:

- **Bible tagged files**: 66 JSON files (one per book), where each English word is linked to its Strong's code (e.g., "beginning" → H7225).
- **Tagged occurrence index**: 14,141 individual Strong's code files containing all 321,621 verse occurrences across the Bible, enabling reverse lookup from any Hebrew or Greek word to every verse where it appears.

A dictionary of 9,333 Hebrew and 5,564 Greek entries was also compiled, providing original-language forms, transliterations, and definitions. This tagging infrastructure powers the Word Donut feature on BibleGo.org (Section 5) and also enabled the TAG retrieval strategy for OT BHT generation (Section 4).

4 Methods

The BibleGoAI pipeline transforms raw multi-author biblical commentary into concise, source-adherent summary paragraphs called *Brief Helpful Texts* (BHTs). The process unfolds across six sequential phases, each building on the output of the previous. Figure 3 illustrates the generation flow.

4.1 Phase 1: Quote Extraction

Given a target verse, the system loads all available commentary from the curated corpus. Each commentator's text is submitted to GPT-3.5-turbo [OpenAI, 2023] with a *choicest prompt* that instructs the model to extract one to three "striking, unique, and enlightening" quotes. The prompt enforces strict constraints:

- Quotes must be **verbatim**—no paraphrasing or injected words.
- Each quote must not exceed 100 words.
- Quotes should capture the most distinctive insights, avoiding generic observations.

Extracted quotes are validated against the original commentary to ensure fidelity. Quotes containing words absent from the source material are flagged and re-extracted. This phase produces a `ChoicestQuotesForVerse` object mapping each commentator to their selected excerpts.

4.2 Phase 2: Retrieval

As an alternative or supplement to direct quote extraction, the system supports a retrieval-augmented approach using dense vector search. Commentary texts are split into chunks using one of three strategies:

1. **Character splitting**: Fixed-size chunks with overlap (via LangChain's `CharacterTextSplitter`) [Chase, 2022].
2. **Recursive splitting**: Hierarchical splitting at paragraph, sentence, and word boundaries

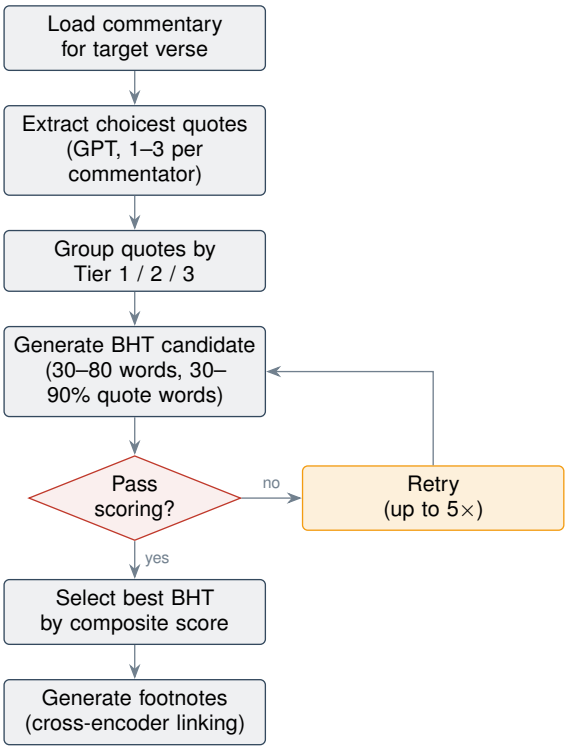

Figure 3: BHT generation flow. Each verse undergoes quote extraction, tiered grouping, constrained generation with up to five retry attempts, quality scoring, and footnote linking.

(`RecursiveCharacterTextSplitter`).

3. **Semantic splitting**: Embedding-aware chunking that respects topical boundaries (`SemanticChunker`).

Chunks are embedded using HuggingFace's `all-mpnet-base-v2` sentence transformer [Song et al., 2020] and indexed with FAISS [Johnson et al., 2019]. At query time, the target Bible verse is embedded and the top-k most semantically similar commentary passages are retrieved, ranked by cosine similarity.

This retrieval mechanism enables coverage of commentaries that span multiple verses, where a single verse may not have a dedicated entry. During OT development, several chunking and retrieval variants were explored—character-based, recursive, semantic, and a tagging-based approach using LLM-driven verse segmentation—to inform the production configuration described below.

4.3 Phase 3: BHT Generation

The core synthesis step produces a BHT from the extracted or retrieved quotes. Commentators are organized into three tiers (Section 3), and the selected quotes are submitted to GPT-3.5-turbo with a *BHT prompt* specifying the following constraints:

- The output must be a single coherent paragraph of 30–80 words in 1–3 sentences.
- Between 30% and 90% of the words must originate from the provided quotes, with a target of roughly 70%.
- The paragraph should draw primarily from Tier 1, sparingly from Tier 2, and minimally from Tier 3.
- No references to commentators by name, verse numbers, or the word "passage."
- The text must not take the form of a list.

These same constraints are enforced at validation time as the hard acceptance gate (Section 4.5). The system makes up to five generation attempts per verse, scoring each candidate and retaining the highest-quality result.

4.4 Phase 4: Split-BHT Variant

To explore whether multi-stage generation improves output quality, a *split-BHT* pipeline decomposes generation into two sub-steps:

1. **Split-A**: An initial prompt produces a draft synthesis focused on extracting key themes from the tiered quotes.
2. **Split-B**: A refinement prompt polishes the Split-A output into a final BHT, enforcing word count, quote proportion, and stylistic constraints.

Both sub-prompts were independently versioned and evaluated. This approach trades additional API calls for potentially higher coherence and fidelity.

4.5 Phase 5: Scoring and Validation

Every generated BHT undergoes a multi-dimensional quality assessment combining hard constraints with continuous metrics.

Hard constraints.

- Word count within 30–80 (strict range: 20–100).
- Quote-word proportion between 30% and 90% (target 70%).
- Absence of forbidden tokens (commentator names, verse references, "passage").
- No list formatting.

Continuous metrics.

- **Commentary accuracy**: Tier-weighted cosine similarity between the BHT and source quotes, computed

via the Universal Sentence Encoder [Cer et al., 2018]. A minimum score of 2.3 is required.

- **Verse accuracy**: Cosine similarity between the BHT and the target verse embedding, measuring topical relevance.
- **Readability**: Flesch-Kincaid Grade Level, Flesch Reading Ease, Gunning Fog Index, Automated Readability Index, and Dale-Chall score [Kincaid et al., 1975, Gunning, 1952].
- **Tier distribution**: The proportion of BHT content attributable to each commentator tier, measured via per-tier semantic similarity.

Among the up to five generation attempts, the BHT with the highest composite commentary-accuracy score is selected. The scoring function prioritizes commentary accuracy while penalizing outputs that violate hard constraints.

4.6 Phase 6: Footnoting and Tagging

The final phase establishes provenance links between the generated BHT and its source commentaries.

Footnoting. Each sentence in the BHT is compared against all source quotes using cross-encoder scoring via `sentence-transformers` [Reimers and Gurevych, 2019]. For each sentence, the system identifies the most similar quote and generates a footnote containing the commentator name, quote index, similarity score, and source location. This enables readers to trace any claim back to its originating commentary.

Commentary tagging. For commentaries spanning multiple verses, a *tag commentary prompt* instructs GPT to segment the text into per-verse sections. The output is a JSON structure mapping each verse number to its relevant content, enabling verse-level retrieval from range-based sources.

4.7 The OT Challenge

The New Testament pipeline described above was designed for commentary that is organized verse-by-verse, which is the norm for NT commentators. After the public release of NT BHTs, the team turned to the Old Testament—and encountered a fundamental structural difference: OT commentary is rarely organized by individual verse. Instead, commentators typically discuss passages spanning multiple verses, making direct quote extraction for a single verse impractical.

Two initial approaches were explored:

1. **Commentary "versification"**: Using GPT to tag multi-verse commentary with per-verse labels, effectively segmenting it into verse-level sections before applying the standard NT pipeline. While conceptually appealing, the accuracy of automated verse tagging proved inconsistent at scale.
2. **Enhanced choicest extraction**: Modifying the choicest prompt to extract verse-specific quotes directly from multi-verse commentary, bypassing the need for segmentation. This also proved unreliable, as the model struggled to isolate verse-relevant content from longer passages.

Both approaches were evaluated over several weeks. The breakthrough came when an AI/ML engineer suggested using retrieval-augmented generation: rather than requiring verse-level segmentation, the system could embed commentary chunks and retrieve the most semantically relevant passages for each verse via vector search. This RAG-based approach (Phase 2) produced usable verse-level BHTs without requiring reliable per-verse segmentation and unlocked OT BHT generation at scale, ultimately enabling completion of the full Bible corpus.

5 The BibleGo.org Platform

The BHTs generated by BibleGoAI are served to end users through **BibleGo.org**, a web application built with Next.js [Vercel, 2023], React, and TypeScript, backed by a PostgreSQL database and hosted on Vercel. The platform currently serves approximately 100–300 unique users per month, with monthly page-view volumes ranging from $\sim$200 to $\sim$2,500 clicks depending on seasonal and community activity, and provides three primary modes of interaction.

Commentary reading. The main reading interface presents a responsive three-column layout (Figure 4): a book selector on the left, the Bible text (NASB) in the center with inline original-language tags, and a commentary panel on the right. When a user selects a verse, the corresponding BHT is displayed along with footnotes that link each sentence to its source commentary. Users can expand individual footnotes to view the original quotes, the commentator attribution, and a similarity score indicating the strength of the connection. Each commentator's full excerpt is also available as a separate

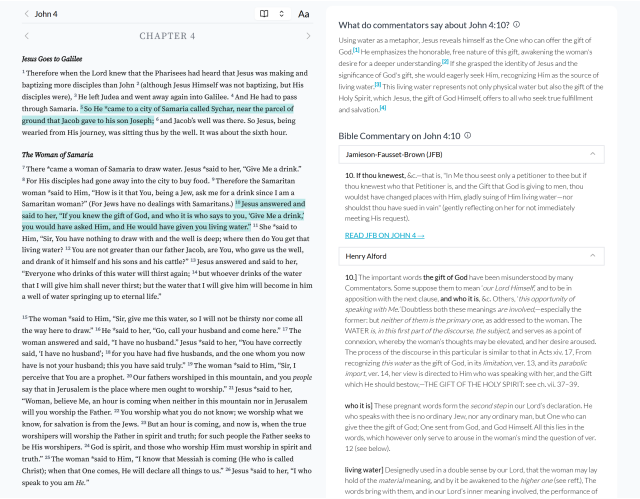

Figure 4: The BibleGo.org desktop reading interface. The three-column layout presents book navigation (left), the NASB Bible text with inline original-language tags (center), and the BHT commentary panel (right) with footnotes linking each sentence to its source commentator.

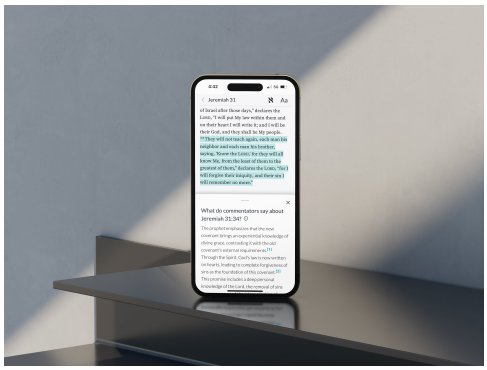

Figure 5: The BibleGo.org mobile experience with the BHT commentary drawer. Footnotes (numbered markers) link each sentence back to its source commentator, allowing users to trace every claim to the original commentary.

card, enabling side-by-side comparison of perspectives. On mobile devices, the commentary panel appears as a slide-up drawer (Figure 5).

Word study. BibleGo.org includes an interactive word study feature—the *Word Donut*—powered by Strong's concordance data built from the original language tagging pipeline (Section 3). Users can select any word in the Bible text to view its original Greek or Hebrew form, transliteration, and definition. An interactive doughnut

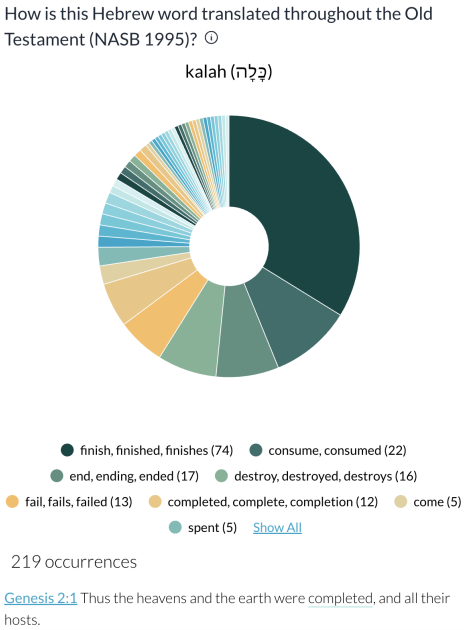

Figure 6: The Word Donut visualization for the Hebrew word *kalah*, showing how it is translated across 219 Old Testament occurrences. Segments represent distinct English renderings (e.g., "finish" 74×, "consume" 22×), enabling users to explore semantic range without knowledge of the original language.

chart then visualizes how the NASB translators chose to render that word across all its occurrences in the Bible, showing each English translation variant and its frequency (Figure 6). This feature supports pagination and filtering by translation variant, enabling original-language exploration without requiring knowledge of Greek or Hebrew.

Full commentary browsing. Users can switch from the BHT summary view to a full commentary mode, which displays the complete chapter-level commentary for any individual commentator (Figure 8). This enables readers who encounter an interesting insight in a BHT to dive deeper into the full context of that commentator's work.

User experience. The application features dark mode support, adjustable font sizes (persisted across sessions), mobile-responsive design with a drawer-based layout for smaller screens, deep linking via URL parameters (e.g., `/read?v=Matthew.1.1`), and an in-app feedback system with structured categories. The UI underwent four rounds of dedicated usability testing—including

Figure 7: Mobile view of the Word Donut for the Hebrew word *racham* ("have compassion") in Isaiah 55. The doughnut chart shows translation frequencies across the Old Testament, enabling word study directly from the reading interface.

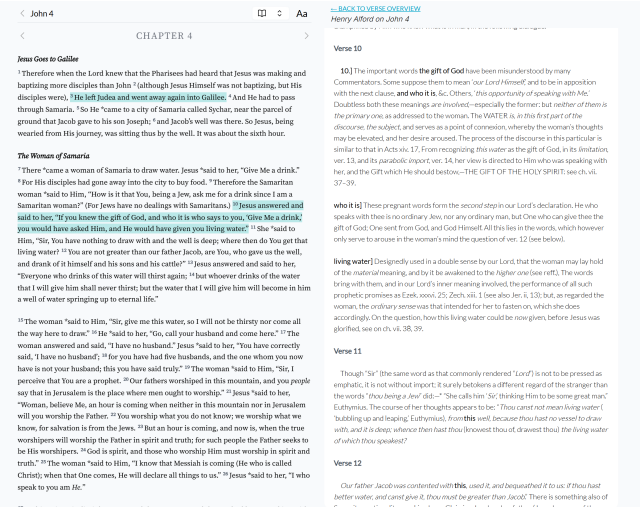

Figure 8: The full commentary browsing mode, displaying a complete chapter-level commentary from an individual commentator. Users can explore the full original text that informs each BHT.

tests focused specifically on the drawer navigation and the word study tool—with feedback directly informing design iterations.

Data flow. Pre-generated BHTs from the BibleGoAI pipeline are loaded into a PostgreSQL database. The Next.js application queries this database through server-side API routes, returning BHT text, footnotes, scoring metadata, and commentator information. Bible text with Strong's number tags is served from static JSON files, while Strong's concordance entries are queried from the database. This hybrid approach balances query flexibility with serving performance.

6 Evaluation

We evaluate BibleGoAI across multiple dimensions: aggregate quality metrics, pipeline comparison, generation attempt analysis, and commentator tier contribution. All metrics are computed from the full corpus of generated BHTs: 7,957 New Testament verses and the full 39-book Old Testament, which is served via two complementary pipelines—a single-pass *regular* pipeline (24 books, 10,492 verses) and a two-stage *split* pipeline (20 books, 12,653 verses) whose two stages (*split-a* and *split-b*) are concatenated into a single served BHT per verse. The two pipelines overlap at the book level on 5 books (1 Kings, Esther, Habakkuk, Hosea, Micah) but cover different verses within them, so the comparison reported below is corpus-wide over each pipeline's own book set rather than a paired per-verse ablation.

6.1 Aggregate Quality Metrics

Table 3 summarizes the key metrics across the NT and OT corpora. The commentary-accuracy score (identical to what is exposed in legacy JSON outputs as `qualityScore`) is a composite tier-weighted similarity between the BHT and its source quotes; quote proportion measures the fraction of BHT words drawn directly from source commentary.

Table 3: Aggregate BHT quality metrics across testaments. Values are mean $\pm$ standard deviation.

Metric	**NT** (n=7,957)	**OT** (n=10,492)
Commentary Accuracy	1.912 ± 0.348	1.785 ± 0.282
Word Count	84.2 ± 11.1	62.9 ± 11.6
Quote Proportion (%)	39.6 ± 7.4	31.8 ± 6.3
Verse Accuracy	—	0.417 ± 0.147

The NT corpus shows higher average commentary-accuracy scores (1.912 vs. 1.785), with longer BHTs (84 vs. 63 words) and higher quote proportions (39.6% vs. 31.8%). These differences are consistent with the structural contrast between the NT pipeline (verse-by-verse commentary, direct per-verse quote extraction) and the OT pipeline (multi-verse commentary, retrieval-augmented generation) described in Section 4. Verse accuracy was computed only for the OT corpus; see below for discussion.

A note on NT word count. The observed NT mean (84.2 words) sits just above the target upper bound (80). The generation loop runs up to five attempts per verse and, if no candidate satisfies the full set of soft constraints (30–80 words, quote-proportion, etc.), the highest-scoring attempt is retained regardless of which constraints it misses. This best-of-five fallback prioritizes commentary quality over strict length compliance, yielding a mean that slightly overshoots the 80-word upper bound; see Section 4 for the full constraint-enforcement protocol.

Verse accuracy (OT only). The *verse accuracy* metric—cosine similarity between the BHT and the target-verse embedding—was computed only for the OT corpus in the reported results. Because the NT and OT pipelines differ (direct per-verse extraction for NT vs. retrieval-augmented generation for OT; see Section 4.7), the OT verse-accuracy value (0.417 ± 0.147) should be read as a sanity check on the OT RAG pipeline rather than as directly comparable to a would-be NT value. Computing this metric retrospectively across the full NT corpus is straightforward and is noted as near-term follow-up work.

6.2 Pipeline Comparison: Regular vs. Split

Two OT generation pipelines were built, each run once over a different subset of the OT canon:

- **Regular pipeline**: A single-pass generation that produces one BHT per verse from tagged commentary, targeting 30–80 words. Run over 24 OT books (10,492 verses), predominantly wisdom, prophetic, and selected historical books where per-verse commentary coverage was adequate.
- **Split pipeline**: A two-stage generation where *split-a* produces a short factual paragraph and *split-b* adds one enriching sentence; the two outputs are concatenated per-verse into a single BHT, with a combined score computed as the mean of the two stage scores. Run over 20 OT books (12,653 verses), predominantly Pentateuch and historical narrative where verse-range commentary dominates and the split was developed to handle broader context.

A note on the comparison. The two pipelines were developed sequentially and run over mostly disjoint book sets. Five books (1 Kings, Esther, Habakkuk, Hosea, Micah) appear in both pipelines' outputs, but inspection reveals that the two pipelines covered *different chapters* of these books with no overlapping verses—the regular-pipeline outputs for these books are experimental partial runs that never reached production. As a result, no paired per-verse head-to-head comparison is possible from the generated corpus; Table 4 reports corpus-wide means over each pipeline's own book set.

Table 4: Pipeline comparison across the OT corpus. Numbers are corpus-wide means over each pipeline's own book set (regular: 24 books; split: 20 books); the two sets overlap at the book level on 5 books but the regular-pipeline outputs for those books are partial experimental runs covering different verses than the split-pipeline outputs, so a paired per-verse comparison is not possible.

Metric	Regular	Split (combined)
n (verses)	10,492	12,653
n (books)	24	20
Commentary Accuracy	1.785	1.822
Word Count	61.4	83.2
Quote Prop. (%)	31.8	32.5

The two pipelines produce BHTs with comparable commentary-accuracy means (1.78 vs. 1.82) and near-identical quote-token proportions ($\sim$32%), indicating comparable source-fidelity behavior. The main observable difference is length: split-pipeline BHTs are roughly 35% longer (83 vs. 61 words) because the concatenation of *split-a* and *split-b* yields a two-paragraph output. Because the two pipelines cover different book sets, these corpus-wide means should be read as per-pipeline performance characteristics on the books each was actually run over—not as a controlled ablation.

Stage-level view (split pipeline only). Within the split pipeline, *split-a* produces longer, higher-scoring paragraphs on its own (mean commentary-accuracy 1.935, mean word count 53.7) while *split-b* produces shorter, lower-scoring enriching sentences (mean 1.710, mean word count 29.5). These are not alternative strategies—both stages always run together, and the combined served BHT is their concatenation—but the per-stage numbers illustrate the intended asymmetric roles of the two sub-prompts.

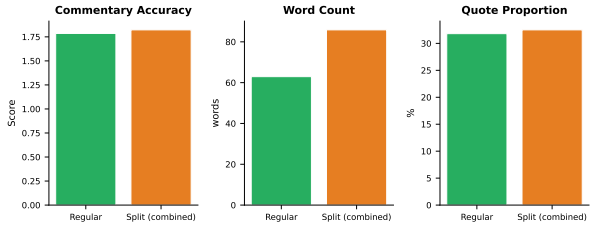

Figure 9: Corpus-wide means for the Regular and Split (combined) OT pipelines across each pipeline's own book set.

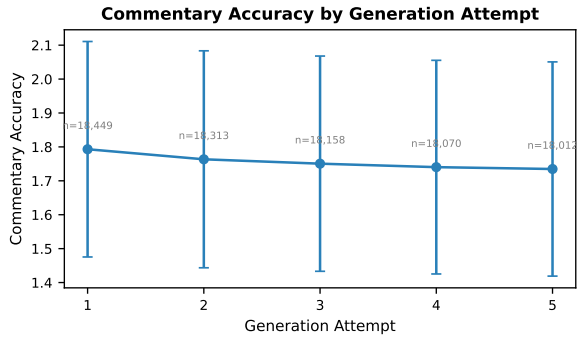

Figure 10: Mean commentary-accuracy score at each generation attempt (1–5). Scores decrease with attempt index because later attempts are only triggered when earlier ones fail the hard constraint gate, so attempts 2–5 are drawn from a harder subset of verses.

6.3 Generation Attempt Analysis

The system generates up to five candidate BHTs per verse and selects the highest-scoring attempt. Figure 10 shows the average commentary-accuracy score at each attempt number across all NT and OT runs combined (n=43,755 first attempts), which reveals a counter-intuitive pattern: mean scores *decrease* monotonically with attempt index (1.74, 1.70, 1.69, 1.68, 1.67 for attempts 1 through 5). This is not evidence that later generations are worse in absolute terms; rather, it reflects the selection dynamics of the loop: later attempts are only triggered when earlier attempts fail the hard constraint gate (word count, quote proportion, forbidden tokens), so attempts 2–5 are drawn from a harder subset of verses where the generator has already struggled. The best-of-five selection then routinely picks the first-attempt BHT. In the production run, the majority of verses succeeded on attempt 1, and the retry budget primarily functioned as safety capacity for the minority of verses where the first attempt tripped a hard constraint.

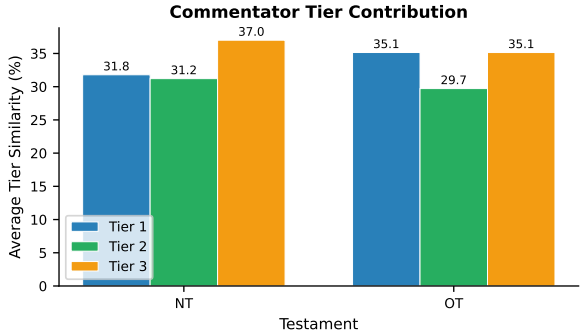

Figure 11: Average commentator-tier contribution to BHT semantic similarity, as a percentage summing to 100% per verse. Tier 3 contributes at or above the level of Tier 1 in both corpora, indicating the tier instruction is not strongly followed in the produced output.

6.4 Commentator Tier Contribution

Figure 11 shows the distribution of tier-weighted semantic similarity across the three commentator tiers, expressed as the percentage of each tier's contribution to the final BHT's similarity profile (values sum to 100% per verse, averaged across the corpus). Both testaments show a flatter distribution than the prompt's "primarily Tier 1, sparingly Tier 2, minimally Tier 3" instruction would suggest. NT: Tier 1 31.8%, Tier 2 31.2%, Tier 3 37.0%. OT: Tier 1 35.1%, Tier 2 29.7%, Tier 3 35.1%. In both corpora the Tier 3 commentators (Calvin, Gill, Wesley, and others) contribute at or above the level of Tier 1, indicating that the tier instruction is not strongly followed in the produced output. One likely cause is that several Tier 3 commentators produce longer and more thematically broad excerpts than the Tier 1 commentators, yielding higher surface-level similarity scores even when they contribute less content by intent. Reconciling the intent-based tiering with similarity-based measurement is a natural follow-up for future tuning.

6.5 Commentary-Accuracy Score Distribution

Figure 12 presents the distribution of commentary-accuracy scores across the NT and OT corpora. Both distributions are approximately Gaussian with a right-skew tail toward higher-scoring verses. For the NT corpus (n=7,957), the 5th, 25th, 50th, 75th, and 95th percentiles are 1.32, 1.68, 1.93, 2.14, and 2.46 respectively. The combined OT corpus (n=35,798 across regular and split pipelines) runs slightly lower: 1.29, 1.58, 1.80, 2.02, and 2.39 at the same percentiles. The minimum-

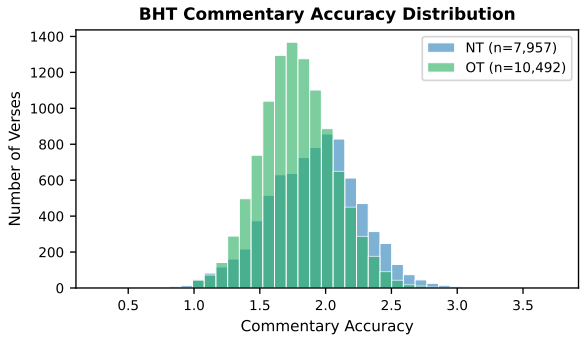

Figure 12: Distribution of commentary-accuracy scores across the NT (n=7,957) and OT (n=35,798, regular and split pipelines combined) corpora. Both distributions are approximately Gaussian with a right-skew tail; the nominal acceptance threshold (2.3) sits near the 85th percentile of produced scores.

acceptance threshold in the hard-constraint gate is 2.3 (Section 4, Phase 5), which sits near the 85th percentile of produced scores—a deliberately permissive threshold reflecting the best-of-five fallback policy: if no attempt clears 2.3, the highest-scoring attempt is still retained to avoid coverage gaps. A meaningful minority of retained BHTs therefore fall below the nominal threshold, a known trade-off between full-Bible coverage and score strictness.

6.6 Human Evaluation

In addition to automated metrics, BHT quality was evaluated through two rounds of structured user testing with members of the project's Bible study community:

- **BHT V2 Quality Test**: Human raters scored NT BHTs on a normalized quality scale. Scores ranged from 50.4 to 94.2 out of 100, with a mean of 67.5. The highest-rated BHT (Romans 6:7, score 94.2) demonstrated strong synthesis of commentary insights, while lower-scoring BHTs guided subsequent prompt refinements.
- **OT BHT V1 Quality Test**: A separate round evaluated the first generation of OT BHTs, providing feedback that informed the comparison of RAG versus tagging strategies.

Table 5 presents the human quality scores from the BHT V2 evaluation. Note that these are a separate metric from the automated *commentary-accuracy* score discussed above: raters scored each BHT on a normalized 0–100 quality scale.

Table 5: Human evaluation scores (BHT V2 Quality Test). Human-assigned 0–100 quality ratings; distinct from the automated commentary-accuracy score in Table 3.

Verse	**Human Score**
Romans 6:7	94.2
John 5:22	80.1
Ephesians 1:22	77.3
Ephesians 6:13	65.5
Galatians 5:22	62.7
John 10:27	57.0
Romans 6:21	54.1
Revelation 3:3	50.4
Mean	**67.7**

These human scores, combined with the automated quality metrics, provided a feedback loop that drove iterative prompt refinement—most notably the May 2024 prompt updates that shifted the choicest extraction focus from "striking and enlightening" to "helpful, explanatory, and illuminating" quotes targeting basic words and facts in each verse.

7 Discussion

Strengths. The multi-stage pipeline design offers several advantages over end-to-end generation, though we note upfront that the quantitative comparison against a modern zero-shot baseline remains future work (see Limitations below). By extracting verbatim quotes before synthesis, the system creates an auditable intermediate representation that can be inspected and validated independently of the final BHT. The tiered commentator system ensures that the most authoritative voices dominate the output, while the constraint enforcement (word count, quote proportion, forbidden phrases) prevents many common failure modes of unconstrained LLM generation.

The iterative prompt refinement process—spanning multiple versions across three prompt families that shipped in the production run—proved essential. Early prompt versions produced BHTs that frequently exceeded word limits, included commentator names, or adopted list formatting. Later versions largely eliminated these issues, demonstrating the value of systematic prompt engineering with quantitative feedback loops.

Limitations. Several limitations warrant discussion. First, the entire BHT corpus was generated using GPT-3.5-turbo, which was among the most capable generally available models at the time of development; no regeneration has been performed on newer models because the pipeline was run once to produce the served corpus, and that one-time run's output quality remained acceptable while inference cost for newer models was substantially higher at the time. A full-scale regeneration and re-evaluation on more recent models remains future work. Second, the quality scoring framework, while multi-dimensional, relies heavily on embedding-based similarity, which may not capture all aspects of theological accuracy or interpretive nuance. Third, the quote-proportion constraint (30–90% of BHT words drawn from source quotes, with a 70% target), while ensuring grounding, can sometimes produce BHTs that read as stitched-together phrases rather than naturally flowing prose. Fourth, we do not yet include a zero-shot LLM baseline—e.g., prompting a frontier model directly with all available commentary for a verse and asking it to summarize—so the marginal value of the multi-stage pipeline over modern end-to-end generation has not been quantitatively isolated.

Source adherence vs. fluency. A fundamental tension exists between source adherence and fluency of the generated text. Higher quote proportions ensure adherence but can reduce coherence; lower proportions allow smoother prose but risk introducing LLM-generated content that may not reflect the commentators' actual positions. Our operating range of 30–90% with a 70% target represents a pragmatic balance, validated through iterative testing and user feedback; the observed corpus-level means (39.6% for NT, 31.8% for OT; see Table 3) satisfy the 30% floor and stay well below the 90% ceiling, but consistently fall short of the 70% target. In practice, produced BHTs paraphrase more than the target specifies; the gap between target and observed behavior is a known limitation and a natural candidate for follow-up tuning.

Prompt engineering insights. The evolution of prompts yielded several practical insights: (1) explicit negative constraints ("do not mention commentators") were more effective than positive-only instructions; (2) specifying exact word count ranges reduced variance more than vague length instructions; (3) the tiering language ("primarily," "sparingly," "minimally") meaningfully influenced the tier distribution in generated outputs; and (4) requesting numbered quotes in the extraction phase significantly improved downstream validation.

8 Conclusion and Future Work

We have presented BibleGoAI, a multi-stage pipeline for synthesizing concise, source-adherent summaries of biblical commentary using large language models, retrieval-augmented generation, and semantic similarity scoring. Run once over a curated corpus of 14 commentators spanning four centuries, the pipeline produced Brief Helpful Texts (BHTs) targeting 30–80 words (observed means 63–84 words across the three generation variants) that are grounded in source material, quality-scored across multiple dimensions, and traceable via automated footnotes.

Our evaluation across 7,957 NT verses and the full 39-book OT corpus demonstrates that the pipeline produces consistent, high-quality summaries with strong commentary accuracy and appropriate tier distribution. The comparison of generation strategies (a regular one-shot pipeline versus a two-stage split pipeline for narrative OT books) provides practical guidance for similar systems. The serving application at BibleGo.org, reaching approximately 100–300 unique users per month with monthly page-view volumes ranging from $\sim$200 to $\sim$2,500 clicks, validates the real-world utility of the approach.

Future work. Several directions remain for future development:

- **Unified OT delivery**: Consolidating the regular and split-pipeline OT corpora into a single served variant per verse, enabling cleaner head-to-head strategy evaluation on shared book sets.
- **Additional commentators**: Incorporating more modern and diverse theological perspectives into the corpus.
- **User studies**: Conducting formal user studies to evaluate BHT quality, usefulness, and theological accuracy as perceived by Bible study practitioners.
- **Enhanced original-language tools**: Expanding the word study features to include Hebrew morphology, cross-references, and parallel passage analysis.
- **Multilingual support**: Generating BHTs in languages beyond English to serve the global Bible study community.

- **Fine-tuned models**: Training domain-specific models on biblical commentary to reduce dependency on general-purpose LLMs and potentially improve both fidelity and cost efficiency.

By demonstrating that LLMs can be effectively constrained, scored, and validated to produce source-adherent scholarly summaries, BibleGoAI contributes both a practical tool for Bible study and a methodological framework applicable to other domains where multi-source synthesis with provenance tracking is required.

Lessons for similar projects. Three patterns from this work generalize beyond biblical commentary. First, an extractive intermediate representation—the *choicest quotes* phase—proved decisive: it gives every downstream stage an auditable artifact and makes source fidelity checkable without re-reading every commentary. Second, a best-of-N loop against a hard constraint gate is a low-cost way to get acceptable coverage from an imperfect generator, at the price of a small fraction of retained outputs that miss the nominal quality threshold; the attempt-level analysis in Section 6 shows this trade-off concretely. Third, a volunteer-driven curation and digitization effort can substantially expand a domain's source corpus beyond what any individual team could assemble—but only when paired with a structured consolidation pipeline that merges human correction back into machine-readable form. Domains with similar characteristics—specialized corpora, community stewardship, and a need for claim-level traceability—may find the overall pattern reusable.

Acknowledgments

We thank the following individuals for their contributions to commentary digitization, project direction, and user testing throughout the development of BibleGoAI and BibleGo.org: Madison Bixler, Carlf Cao, Derzan Chiang, Megan Dobos, Austin Funk, Lily Funk, Salomi Gandra, Linsu Han, Algernon Lee, Ben Sheu, Hannah Sprau, Lulu Sun, Josh Wang, and Angela Xia, along with the broader community of volunteers whose work supported the Alford and Govett digitization efforts.

References

Daniel Cer, Yinfei Yang, Sheng-yi Kong, Nan Hua, Nicole Limtiaco, Rhomni St. John, Noah Constant, Mario Guajardo-Cespedes, Steve Yuan, Chris Tar, et al. Universal sentence encoder. In *Proceedings of the 2018 Conference on Empirical Methods in Natural Language Processing: System Demonstrations*, pages 169–174, 2018.

Harrison Chase. Langchain. https://github.com/langchain-ai/langchain, 2022.

Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. BERT: Pre-training of deep bidirectional transformers for language understanding. *Proceedings of NAACL-HLT*, pages 4171–4186, 2019.

Robert Gunning. *The Technique of Clear Writing*. McGraw-Hill, 1952.

Jeff Johnson, Matthijs Douze, and Hervé Jégou. Billion-scale similarity search with GPUs. *IEEE Transactions on Big Data*, 7(3):535–547, 2019.

J. Peter Kincaid, Robert P. Fishburne, Richard L. Rogers, and Brad S. Chissom. Derivation of new readability formulas for navy enlisted personnel. *Research Branch Report 8-75, Naval Technical Training Command*, 1975.

Patrick Lewis, Ethan Perez, Aleksandra Piktus, Fabio Petroni, Vladimir Karpukhin, Naman Goyal, Heinrich Küttler, Mike Lewis, Wen-tau Yih, Tim Rocktäschel, et al. Retrieval-augmented generation for knowledge-intensive NLP tasks. In *Advances in Neural Information Processing Systems*, volume 33, pages 9459–9474, 2020.

NLLB Team. No language left behind: Scaling human-centered machine translation. In *arXiv preprint arXiv:2207.04672*, 2022.

OpenAI. GPT-3.5 turbo. https://platform.openai.com/docs/models/gpt-3-5-turbo, 2023.

Nils Reimers and Iryna Gurevych. Sentence-BERT: Sentence embeddings using siamese BERT-networks. In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing*, pages 3982–3992, 2019.

Kaitao Song, Xu Tan, Tao Qin, Jianfeng Lu, and Tie-Yan Liu. MPNet: Masked and permuted pre-training for language understanding. In *Advances in Neural Information Processing Systems*, volume 33, pages 16857–16867, 2020.

Ashwin Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N. Gomez, Łukasz Kaiser, and Illia Polosukhin. Attention is all you need. *Advances in Neural Information Processing Systems*, 30, 2017.

Vercel. Next.js: The react framework for the web. https://nextjs.org/, 2023.